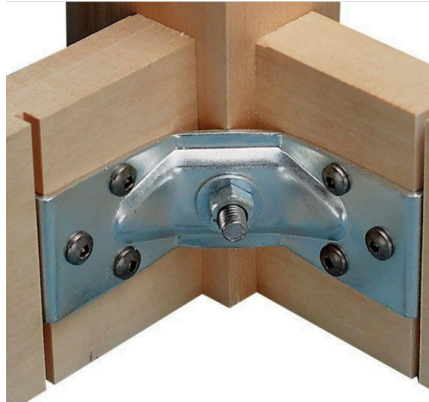


Element B

Prior Design Attempt #1 - Basic corner table brace

<https://www.woodcraft.com/products/hafele-table-leg-corner-brace?via=573621f569702d06760016d3>

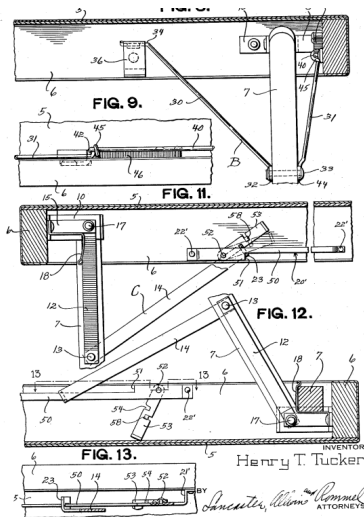


The brace is what Mr L currently uses on the tables. These braces are likely to fall as the screw ends up unscrewing, damaging the stability of the table.

As seen in the table in Mr. L's room, the screws do indeed unscrew under extensive weight. The design is not intended to be a final solution to a weight over 500 pounds. The overall structure of the brace falls short, with its bottle neck being the screws, and their tendency to unscrew after being screwed into the wood.

Prior Design Attempt #2 - Folding table brace

<https://patentimages.storage.googleapis.com/f1/48/8e/199428fcc5a57a/US1940569.pdf>



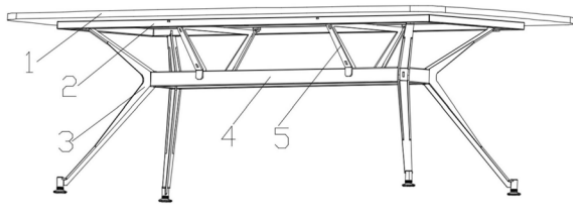
The three figures show a typical folding brace mechanism seen often in folding tables. The folding mechanism is metal with the rest of the table typically being plastic and already molded to proper proportions.

This however would not be perfectly effective as Mr. L doesn't want the legs to fold. The craving into the table itself would inhibit the table from being easily detachable. Additionally, the table would be too expensive, with us having to order metal braces. We can utilize the idea of diagonal braces on the table which would improve stability on the table itself.

Prior Design Attempt #3 - Middle beam table

<https://patentimages.storage.googleapis.com/f5/9f/75/f2646d1fc4f2a5/CN113508986A.pdf>

The table has 5 beams connecting to a middle bar that distributes the weight evenly. This allows for the unconventional format of the 4 legged design.

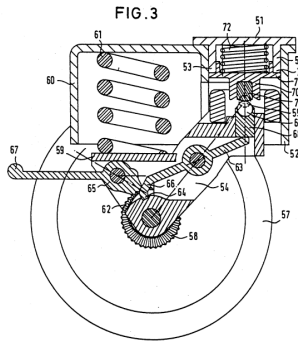


While the table's design in Mr. L has a different set of legs, but we can use the brace design. The spread of the load will improve stability overall. If necessary, we can utilize the design of the legs as well, however, that will be more costly. It will require an entirely different set of legs.

Prior Design Attempt #4 - Lockable Caster Wheel

<https://patentimages.storage.googleapis.com/2c/d6/88/b15e0c03562242/US5014391.pdf>

Image



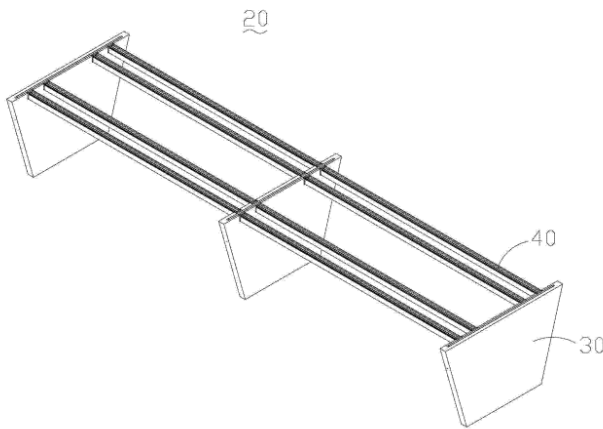
The wheel has a locking mechanism. This wheel would be perfectly positioned on the bottom of a table. The mechanism works by creating enough friction that the wheel is unable to rotate.

The wheel might be a good addition to the table design. However, the issue faced with the locked wheels is they still create a small amount of instability within the design. The wheel would need to be detachable in order to be viable for a stable table design.

Prior Design Attempt #5 - Table top brace

<https://patentimages.storage.googleapis.com/07/cf/fa/bb00f981bd647f/CN103445492A.pdf>

The bar will help support the table top, and by that, allowing the legs to be more stable. This will overall increase durability.



The table bases only harm would be being too expensive. It seems to overall increase the durability to my eyes. The brace would help stabilize the table top but this would also help stabilize the legs, increasing the total weight the table could hold. This would enable students to sit comfortably on top of the table.